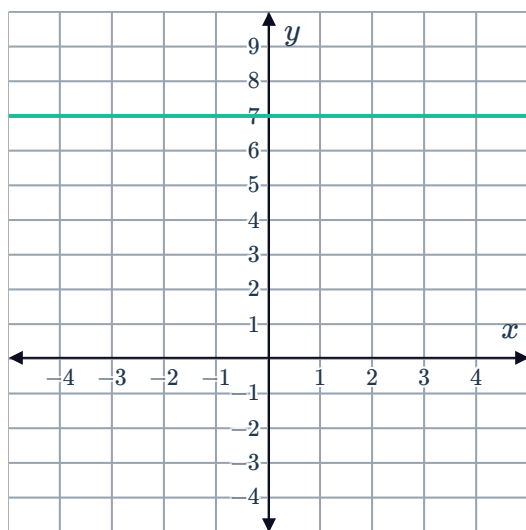


Gradient of a line

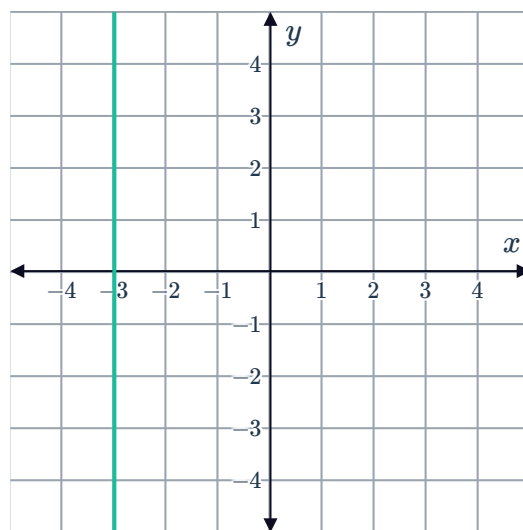


1 Describe the gradient of the lines in the following graphs:

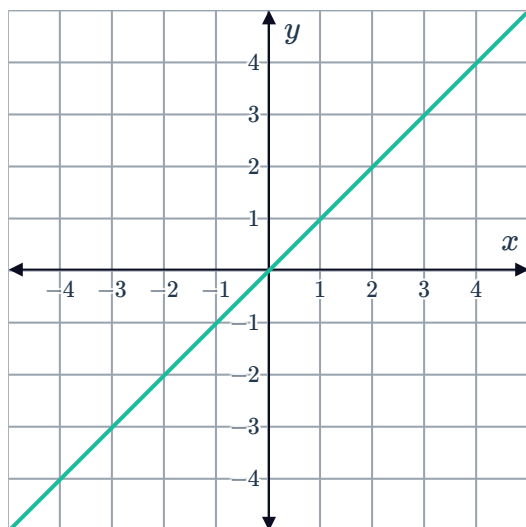
a



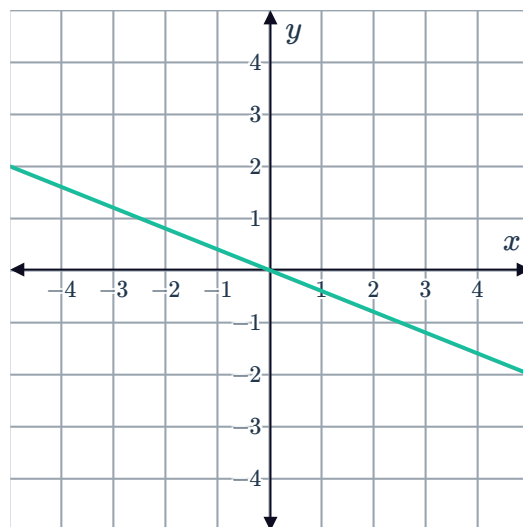
b



c



d



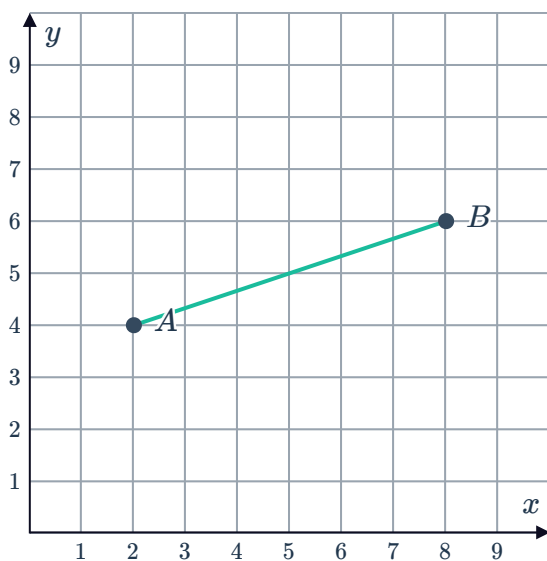
2 For each of the following intervals between points A and B :

i Find the rise.

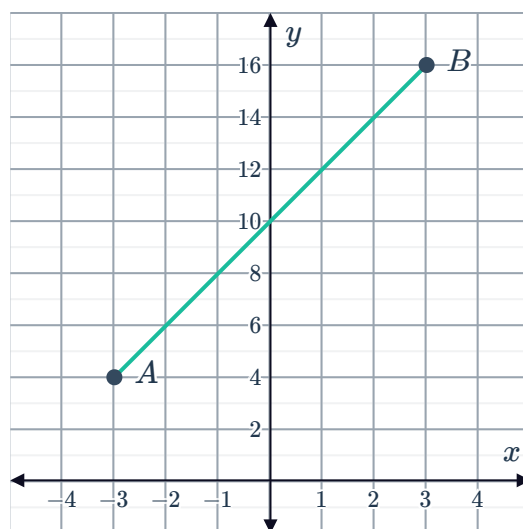
ii Find the run.

iii Find the gradient.

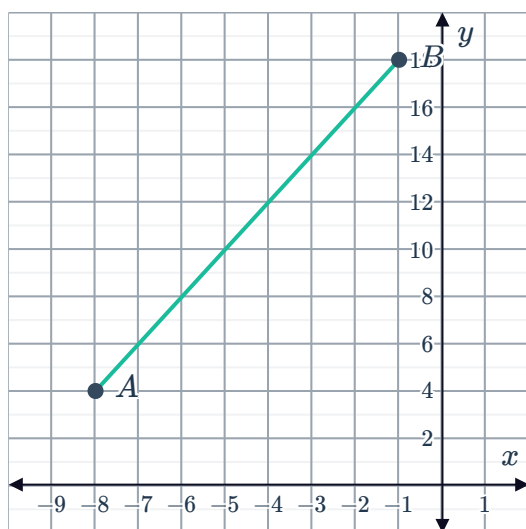
a $A(2, 4)$ and $B(8, 6)$



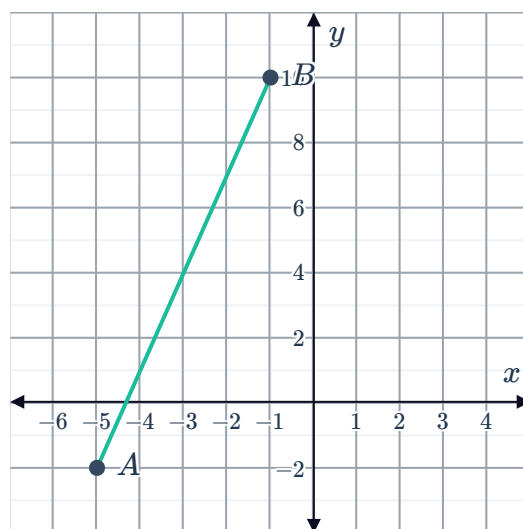
b $A(-3, 4)$ and $B(3, 16)$



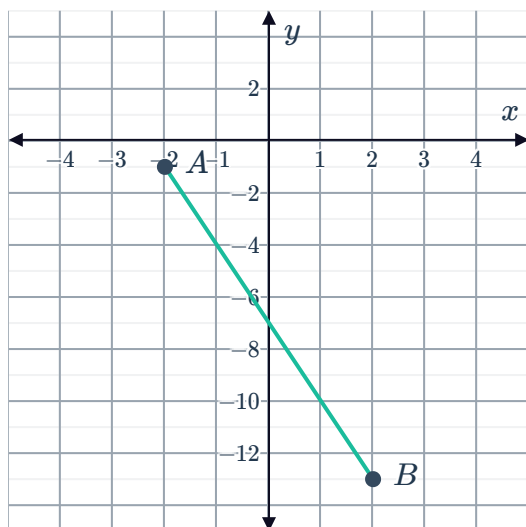
c $A(-8, 4)$ and $B(-1, 18)$



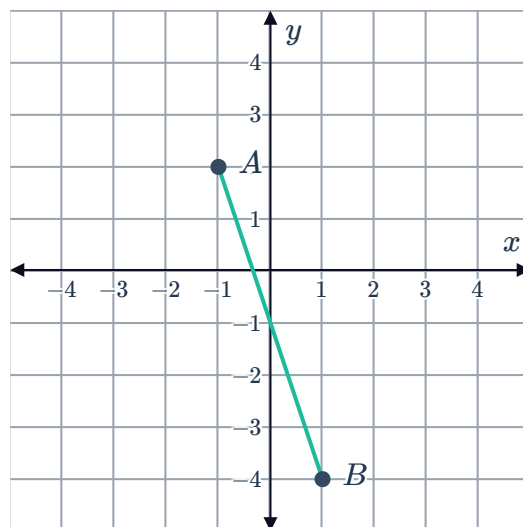
d $A(-5, -2)$ and $B(-1, 10)$



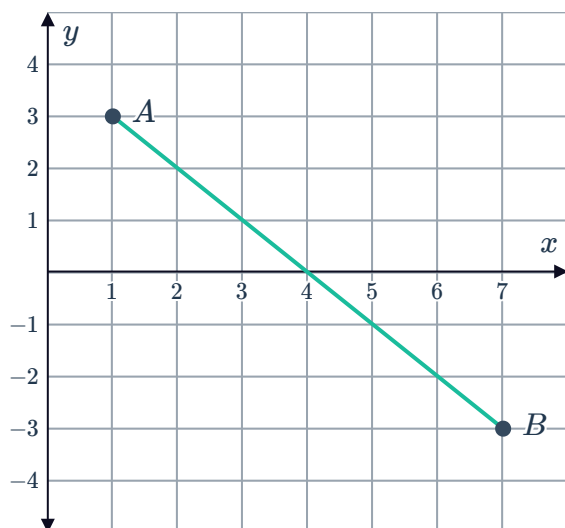
e $A(-2, -1)$ and $B(2, -13)$



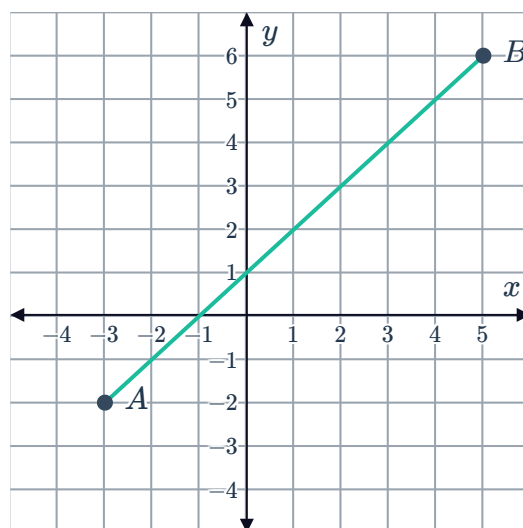
f $A(-1, 2)$ and $B(1, -4)$



g $A(1, 3)$ and $B(7, -3)$



h $A(-3, -2)$ and $B(5, 6)$



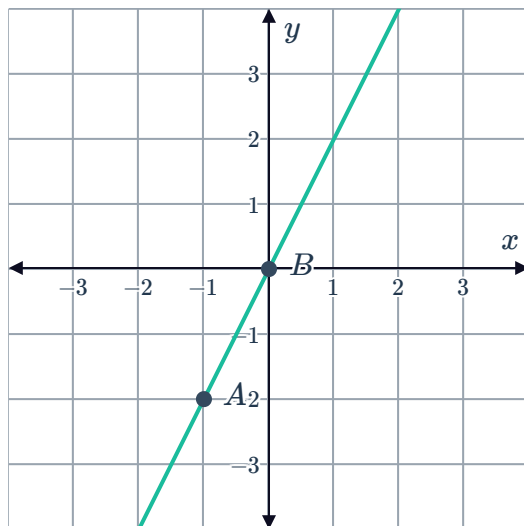
3 Explain why a vertical line has an undefined gradient.

4 State the gradient of any line parallel to the x -axis.

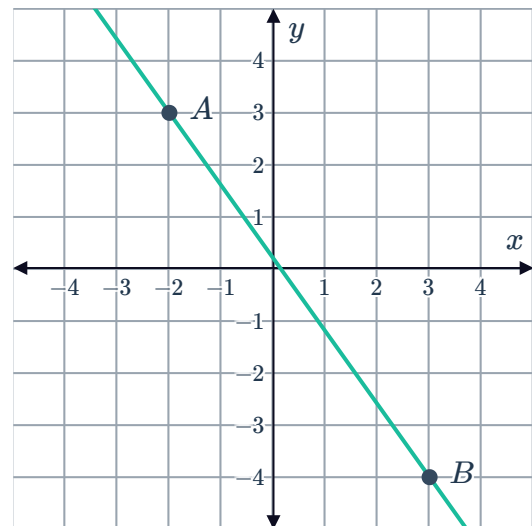
5 Find the gradient of the following lines passing through points A and B :



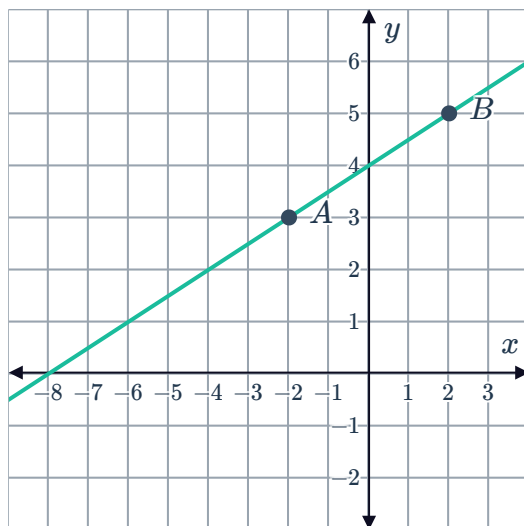
a



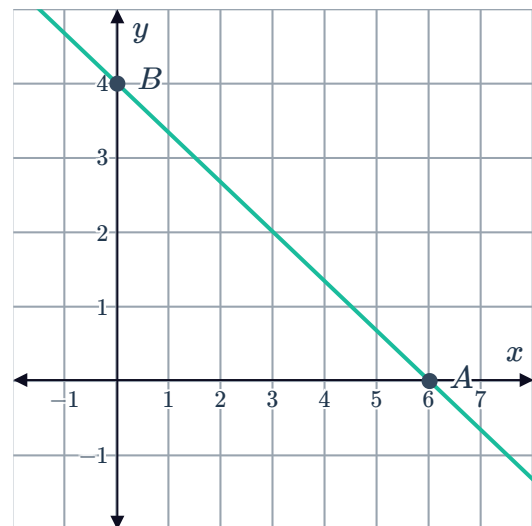
b



c



d

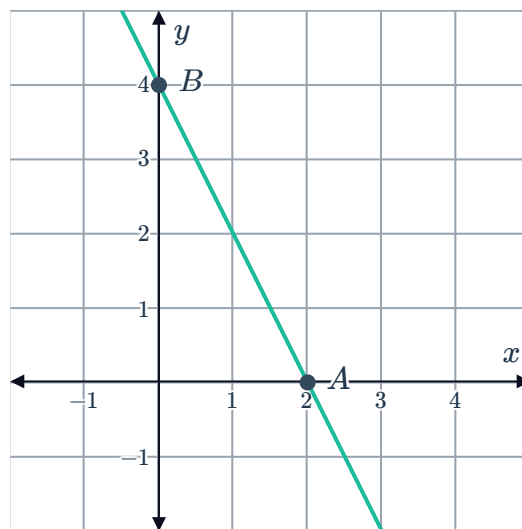


Gradient formula (Extended)

6 Find the gradient of the interval joining $A(-9, 4)$ and $B(-3, -5)$.

7 If we have two points and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$, does it matter which point is (x_1, y_1) and which point is (x_2, y_2) ?

- 8 Consider the line plotted, where $A(2, 0)$ and $B(0, 4)$ both lie on the line.



- a Solve for the gradient of the line.
- b As x increases, what happens to the value of y ?

- 9 Find the gradient of the line that passes through the given points:

- | | |
|------------------------------|----------------------------|
| a $(-1, 0)$ and $(0, 3)$ | b $(-4, 7)$ and $(1, 10)$ |
| c $(1, -4)$ and the origin | d $(2, -6)$ and the origin |
| e $(6, 4)$ and $(3, 4)$ | f $(-6, 5)$ and $(4, 5)$ |
| g $(-2, -5)$ and $(-9, -12)$ | h $(-3, -1)$ and $(-5, 1)$ |

- 10 Consider the points $A(-11, -9)$, $B(-5, 1)$ and $C(-2, 6)$:

- a Find the gradient of AB .
- b Find the gradient of BC .
- c Do the points A , B and C lie in a straight line?

- 11 Consider the following points: $A(26, m - 24)$, $B(-1, m)$ and $C(-10, 9)$.

Find m , given that A , B and C are collinear.

- 12 Given the gradient of the line passing through the two points, find the value of the pronumeral:

- | | |
|--|---|
| a $(4, -3)$ and $(1, t)$, gradient $= -2$ | b $(5, 3)$ and $(2, t)$, gradient $= -4$ |
| c $(5, 3)$ and $(d, 63)$, gradient $= 4$ | d $(11, c)$ and $(-20, 16)$, gradient $= -\frac{4}{7}$ |

- 13 The line $x = 4$ intersects the line $y = 2x - 10$ at the point Q .

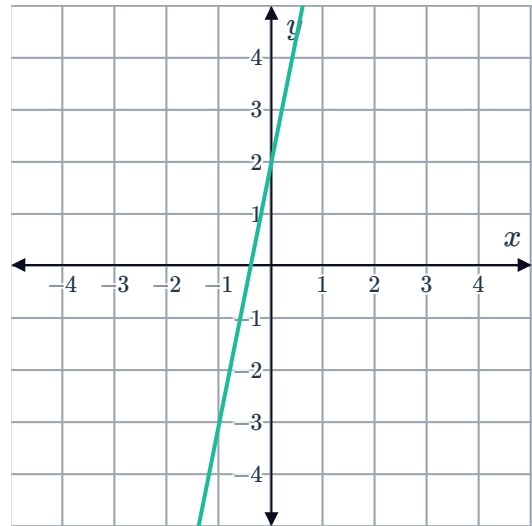
The line $x = -4$ intersects the line $y = 2x - 10$ at point R .

- a Find the coordinates of Q .
- b Find the coordinates of R .



14 Consider the line $y = 5x + 2$ graphed:

- a Find the y -value of the point on the line where $x = 5$.
- b Find the gradient of the line.

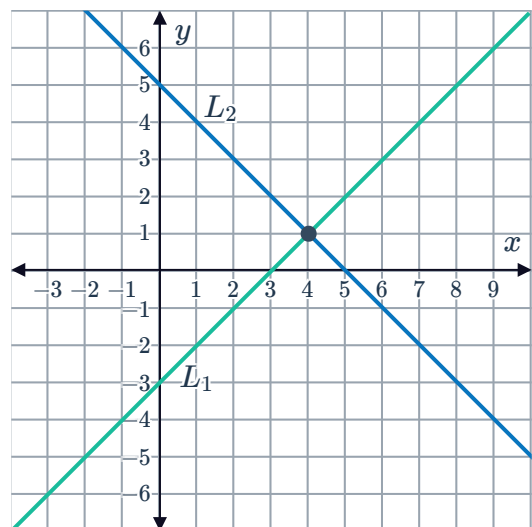


15 Consider the line $y = -2x + 8$.

- a Find:
 - i y -intercept
 - ii x -intercept
- b Sketch the line on a number plane.
- c Hence, find the gradient of the line.

16 Two lines L_1 and L_2 have equations $y = x - 3$ and $y = -x + 5$ respectively. The lines and their point of intersection have been graphed:

- a When $x = 4$, find the y -coordinate of the corresponding point on L_1 .
- b When $x = 4$, find the y -coordinate of the corresponding point on L_2 .
- c Find the gradient of the two lines:
 - i L_1
 - ii L_2
- d Find the product of their gradients.





Properties of polygons (Extended)

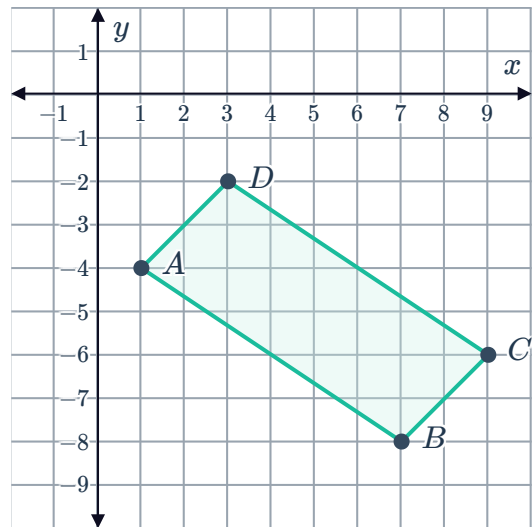
17 Consider the quadrilateral $ABCD$ that has been graphed on the number plane:

a Find the gradient of following sides:

i AB ii CD

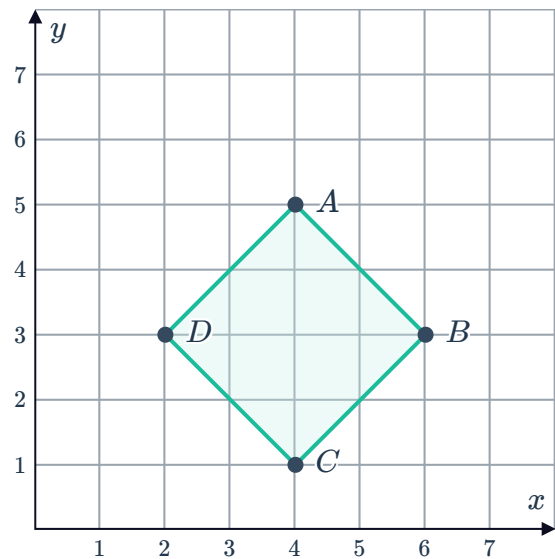
iii AD iv BC

b What type of quadrilateral is $ABCD$?
Explain your answer.



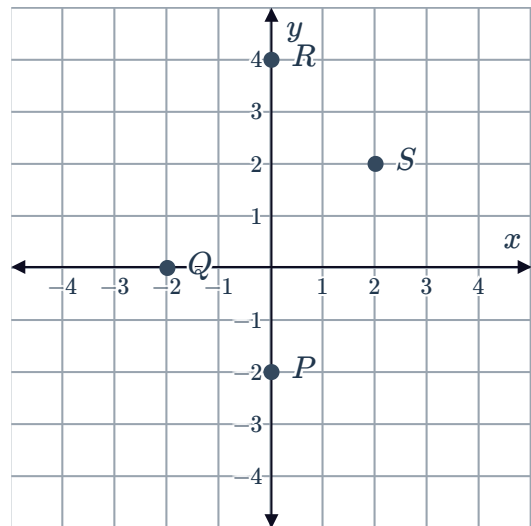
18 The 4 vertices of square $ABCD$ have been plotted on a number plane.

- a** Find the gradient of side AB .
- b** Find the gradient of side BC .
- c** Find the product of the gradients in parts (a) and (b).
- d** If two lines are perpendicular their gradients multiply to -1 . Are sides AB and BC perpendicular?



19 The points $P(0, -2)$, $Q(-2, 0)$, $R(0, 4)$ and $S(2, 2)$ are graphed below:

- Find the gradient of PQ .
- Find the gradient of RS .
- Are PQ and RS parallel?
- Are QR and PS parallel?
- Identify the type of quadrilateral $PQRS$ is.



20 The vertices of $\triangle ABC$ are $A(9, -12)$, $B(4, 4)$ and $C(-8, -5)$. The sides AB and AC have midpoints D and E respectively.

- Find the coordinates of points D and E .
- Find the gradient of side BC .
- Find the gradient of side DE .
- Are BC and DE parallel to each other?

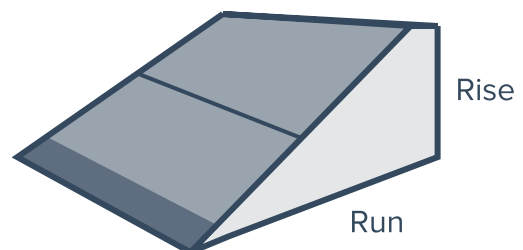
21 $A(-4, -2)$, $B(2, 1)$ and $C(2, -4)$ are the vertices of a triangle.

- Name the side of the triangle that is a vertical line.
- Find the area of the triangle.

Applications

22 Consider the following ramp:

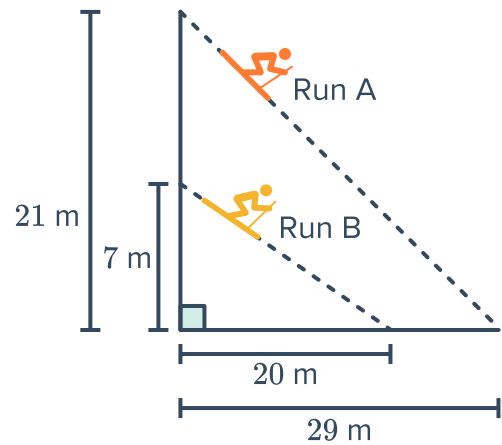
- Find the gradient of this skateboard ramp if it rises 0.9 metres above the ground and runs 1 metre horizontally at the base.
- The ramp can only be used as a 'beginner's ramp' if for every 1 metre horizontal run, it has a rise of at most 0.5 metres. Can it be used as a 'beginner's ramp'?



23 A certain ski resort has two ski runs as shown in the diagram:



- a Find the gradient of Run A. Round your answer to two decimal places.
- b Find the gradient of ski run B. Round your answer to two decimal places.
- c Which run is steeper?



24 A paratrooper falls to the ground along a diagonal line. His fall begins 1157 m above the ground, and the line he follows has a gradient of 1.3. That is, he falls 1.3 m vertically for every 1 m he moves across horizontally.

How far horizontally across the ground does he land from his initial position in the sky?